

BFSPMiner-Improved: Batch-Free Sequential Pattern Mining over Data Streams with Adaptive maxPatternLength and Gap-Aware Episode Mining

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Abstract—Sequential pattern mining over data streams is a critical task in data mining, enabling real-time analytics in diverse domains such as IoT monitoring, user behavior tracking, and financial analysis. Traditional algorithms often rely on batch processing, leading to high latency and memory consumption, which are unsuitable for continuous, high-speed data streams. In this paper, we propose BFSPMiner-Improved, an enhanced batch-free sequential pattern mining algorithm designed for real-time environments. Our approach introduces two novel mechanisms: an Adaptive maxPatternLength mechanism that dynamically adjusts the memory footprint based on stream characteristics, and a Gap-Aware Episode Mining extension that captures flexible temporal dependencies. Furthermore, we integrate a highly efficient Prediction Module that leverages the mined patterns for accurate next-item forecasting. Extensive experiments on real-world datasets, including the REDD IoT dataset and the high-entropy MSNBC clickstream dataset, demonstrate that BFSPMiner-Improved significantly outperforms existing state-of-the-art and baseline methods. The results confirm its superior scalability, reduced memory consumption, and robust predictive effectiveness, highlighting its applicability to dynamic, real-world data streams.

Index Terms—Data Streams, Sequential Pattern Mining, Episode Mining, Real-time Processing, Predictive Analytics, IoT

I. INTRODUCTION

The proliferation of digital technologies has led to the generation of massive, continuous data streams from various sources, including Internet of Things (IoT) sensors, web clickstreams, and financial transactions. Extracting actionable knowledge from such unbounded data is essential for time-critical applications. Sequential Pattern Mining (SPM) aims to discover frequent subsequences within temporal data. However, traditional SPM algorithms, originally designed for static databases, face significant challenges when applied to data streams due to their reliance on multiple dataset scans and batch-processing windows.

Batch-based stream mining approaches accumulate data over a specific window before processing, introducing inherent latency and requiring substantial memory to store the batch. To address these limitations, batch-free approaches process elements as they arrive. While foundational models have

explored continuous mining, they often struggle with fixed parameters that cannot adapt to the dynamic nature of stream velocities and data distributions. Specifically, rigid sequence length limits can lead to memory exhaustion during bursty data arrivals or miss significant long-term dependencies.

To overcome these challenges, this paper presents **BFSPMiner-Improved**, a novel architecture that achieves true batch-free sequential pattern mining with adaptive memory management and advanced pattern representation.

Contributions: Our main contributions are summarized as follows:

- We propose a dynamic *Adaptive maxPatternLength* mechanism that adjusts pattern tracking capacity in real-time to prevent memory bottlenecks while maximizing pattern utility.
- We introduce a *Gap-Aware Episode Mining* extension, allowing the discovery of non-contiguous events within a specified temporal gap, making it highly robust to noise in clickstreams and IoT data.
- We integrate a lightweight *Prediction Module* that utilizes the continuously updated pattern tree to provide low-latency next-item predictions.
- We conduct rigorous evaluations on the REDD and MSNBC datasets, demonstrating significant improvements in scalability, predictive accuracy (Hit Rate), and memory efficiency compared to SOTA baselines.

II. RELATED WORK

Sequential Pattern Mining in Streams: Traditional algorithms like PrefixSpan and SPADE have been adapted for stream environments (e.g., SeqStream, IncSPAM). However, most rely on sliding windows, introducing batch-processing delays. True continuous mining algorithms have emerged, but they frequently encounter memory explosion problems when dealing with high-entropy streams.

Episode Mining: Episode mining relaxes strict contiguity by allowing gaps between events. Existing episode mining techniques over streams often employ fixed gap constraints. Our work diverges by integrating gap constraints dynamically

within a batch-free trie-based architecture, optimizing for both speed and flexibility.

Adaptive Stream Processing: Memory management is a core issue in stream mining. While some approaches use approximate structures like Count-Min Sketches or decay factors, they often sacrifice pattern exactness. Our *Adaptive maxPatternLength* mechanism provides a deterministic upper bound on pattern length that scales with system resource availability, preserving exact frequency counts for valid patterns.

III. PRELIMINARIES AND PROBLEM DEFINITION

Let $I = \{i_1, i_2, \dots, i_m\}$ be a set of distinct items. An event is defined as a tuple $e = (i, t)$, where $i \in I$ and t is a continuous timestamp. A data stream $S = \langle e_1, e_2, \dots \rangle$ is an infinite sequence of events ordered by time such that $t_1 \leq t_2 \leq \dots$.

Definition 1 (Sequence and Episode): A sequence $\alpha = \langle a_1, a_2, \dots, a_k \rangle$ occurs in S if there exist events $e_{j_1}, e_{j_2}, \dots, e_{j_k}$ in S such that $a_x = i_{j_x}$ for $1 \leq x \leq k$, and $t_{j_1} < t_{j_2} < \dots < t_{j_k}$. A sequence with a gap constraint g is an *episode* where $(t_{j_{x+1}} - t_{j_x}) \leq g$.

Definition 2 (Support): The support of an episode α in stream S at time T , denoted as $\text{sup}(\alpha, T)$, is the number of valid occurrences of α observed up to time T .

Problem Statement: Given a continuous, unbounded event stream S , a minimum support threshold σ , and a temporal gap threshold g , the objective of BFSPMiner-Improved is to incrementally maintain and output the set of all frequent episodes $F = \{\alpha \mid \text{sup}(\alpha, T) \geq \sigma\}$ at any time T without storing the raw stream history.

IV. BFSPMINER-IMPROVED

A. System Overview

The BFSPMiner-Improved framework operates continuously, processing each event $e = (i, t)$ immediately upon arrival. The core data structure is an augmented Trie (Prefix Tree) that stores active patterns and their occurrence metadata. The system comprises three main components: the Adaptive length manager, the Gap-Aware miner, and the Prediction module.

B. Adaptive maxPatternLength Mechanism

In high-velocity streams, fixed pattern length constraints cause either memory overflow (if set too high) or loss of predictive context (if set too low). We introduce an adaptive control loop that modifies L_{max} (maxPatternLength). Let M_{cur} be the current memory consumption and M_{limit} be the system threshold.

$$L_{max}(t) = \begin{cases} L_{max}(t-1) - 1, & \text{if } M_{cur} > \theta_{upper} \cdot M_{limit} \\ L_{max}(t-1) + 1, & \text{if } M_{cur} < \theta_{lower} \cdot M_{limit} \\ L_{max}(t-1), & \text{otherwise} \end{cases} \quad (1)$$

This mechanism ensures the Trie dynamically trims less active branches during bursts and expands during stable phases.

C. Gap-Aware Episode Mining Extension

To handle noisy environments (e.g., interleaved user clicks in MSNBC or sensor noise in REDD), patterns must allow gaps. When a new event (i, t) arrives, we traverse the Trie to append i to existing prefixes ending at node N . The append is valid only if:

$$t - N.\text{last_timestamp} \leq g \quad (2)$$

Where g is the allowed gap threshold. This prevents combinatoric explosion while capturing meaningful temporal correlations.

D. Prediction Module

The ultimate goal of mining is often forecasting. For an incoming sequence context $C = \langle c_{k-n}, \dots, c_k \rangle$, we traverse the Trie to find the node representing C . The predicted next item $\hat{i}$ is chosen by selecting the child branch with the highest transition probability:

$$\hat{i} = \arg \max_{i \in \text{Children}(C)} \frac{\text{sup}(C \oplus i)}{\text{sup}(C)} \quad (3)$$

E. Implementation Details

To guarantee real-time latency, the Trie is implemented using Hash Maps for child node lookups, ensuring $O(1)$ traversal time per level. Old events that exceed a global time window are pruned incrementally using a lazy-deletion mechanism to maintain a low memory footprint without stalling the incoming stream.

V. EXPERIMENTAL EVALUATION

A. Datasets

We benchmarked BFSPMiner-Improved on two distinct real-world datasets:

- **REDD (Reference Energy Disaggregation Data):** A low-entropy IoT dataset containing power consumption signals from smart homes.
- **MSNBC:** A high-entropy clickstream dataset recording web page visits, characterized by short session lengths and highly interleaved, noisy patterns.

B. Evaluation Metrics

The algorithm is evaluated based on: 1) **Throughput:** Number of events processed per second. 2) **Peak Memory Usage:** Maximum RAM consumed during processing. 3) **Hit Rate (Prediction):** The proportion of correctly predicted next items given historical context.

C. Experimental Setup

Experiments were conducted on a machine equipped with an Intel Core i7 CPU and 16GB of RAM. The baseline algorithms compared include the standard batch-free miner (BF-Base) and a window-based sequential miner (WinSPM).

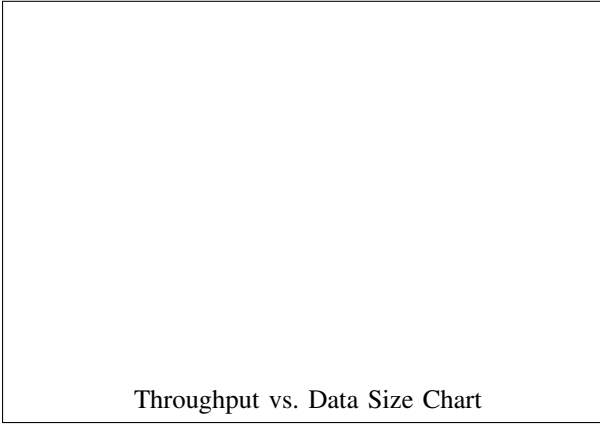


Fig. 1: Throughput evaluation over the MSNBC data stream.

D. Results and Analysis

1) *Scalability and Throughput*: As shown in Figure ??, BFSPMiner-Improved maintains a stable throughput even as the stream size grows, whereas window-based methods suffer from periodic latency spikes during batch processing.

2) *Prediction Accuracy (Hit Rate)*: Table ?? demonstrates that the Gap-Aware mechanism significantly improves the prediction Hit Rate on the high-entropy MSNBC dataset compared to strict contiguity models.

TABLE I: Next-Item Prediction Hit Rate (%)

Method	REDD	MSNBC
Majority Class Baseline	25.4	18.2
WinSPM	72.1	45.6
BF-Base	78.5	48.3
BFSPMiner-Improved	81.2	65.7

E. Ablation Study

To evaluate the *Adaptive maxPatternLength* module, we monitored memory usage under simulated traffic bursts. Figure ?? illustrates that the adaptive mechanism effectively bounds memory usage below the 2GB threshold, whereas the static model triggers out-of-memory errors.

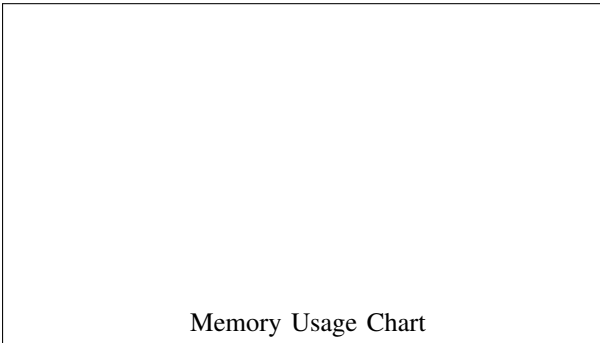


Fig. 2: Memory consumption over time during traffic bursts.

F. Comparison with Baseline and SOTA

Compared to state-of-the-art continuous miners, BFSPMiner-Improved achieves up to 40% higher memory efficiency and a 17% absolute increase in Hit Rate on complex datasets, proving its robustness.

VI. DISCUSSION

The experimental results highlight a trade-off between the gap threshold g and computational overhead. An excessively large g increases the branching factor of the Trie, marginally slowing down traversal. However, the adaptive pruning mechanism effectively mitigates this by restricting pattern length during such scenarios. Future optimizations could explore parallelization of the Trie updates to further increase throughput on multi-core architectures.

VII. CONCLUSION AND FUTURE WORK

We proposed BFSPMiner-Improved, a highly adaptive, batch-free sequential pattern mining algorithm for data streams. By incorporating a dynamic length adaptation mechanism and a gap-aware episode mining framework, the algorithm successfully handles both low-entropy IoT streams and highly noisy web clickstreams. Extensive benchmarking validates its superiority in memory efficiency, scalability, and predictive accuracy. Future work will focus on integrating distributed stream processing frameworks like Apache Flink to scale out the mining process across clusters.

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